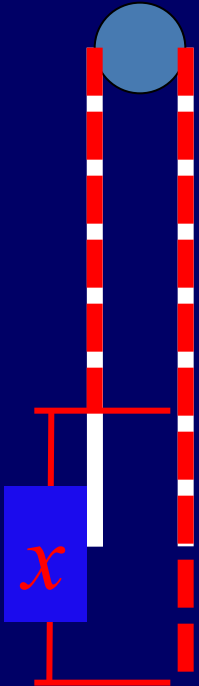
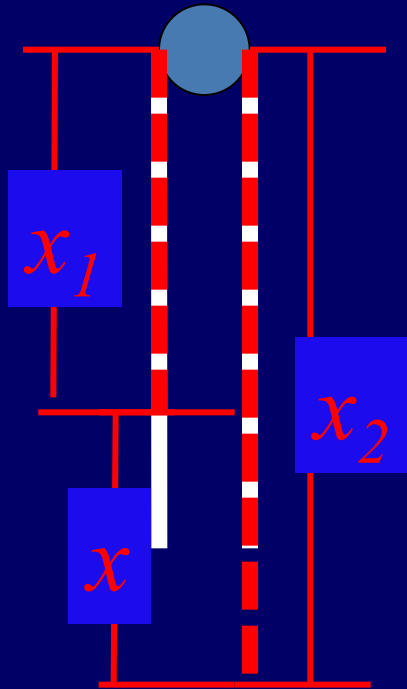


Quiz 7. A uniform flexible chain of length l with linear density $\lambda = m/l$ passes over a pulley with the radius R and the mass M . If there is no relative motion between the chain and the pulley. Initially, there are the same lengths in two sides of the pulley. If we give a very small downward drag force of one end of the chain, calculate its acceleration speed when the difference of two ends of the chain is x .



Answer1: $a = R\alpha$



$$x_2 \lambda g - T_2 = x_2 \lambda a$$

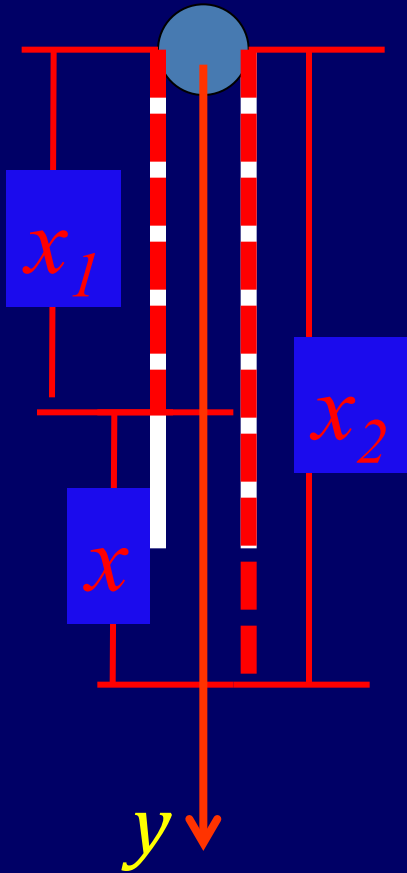
$$T_1 - x_1 \lambda g = x_1 \lambda a$$

$$(T_2 - T_1)R = \left(\frac{1}{2}MR^2 + \pi R\lambda R^2\right)\alpha$$

$$a = \frac{xmg}{(m + M / 2)l}$$

$$l = x_1 + x_2 + \pi R$$

Answer2: $l=L+\pi R$ $L=x_1+x_2$ $y_{c0}=L/4$



$$m_1 = x_1 \lambda \quad y_1 = x_1 / 2$$

$$m_2 = x_2 \lambda \quad y_2 = x_2 / 2$$

$$y_c = (m_1 y_1 + m_2 y_2) / (m_1 + m_2) = (x_2^2 + x_1^2) / 2L$$

$$\frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 = L \lambda g (y_c - y_{c0}) = \frac{g \lambda}{2} (x_2^2 + x_1^2 - \frac{L^2}{2})$$

$$m a + I \alpha \omega / v = g \lambda (x_2 - x_1)$$

$$v = \omega R, \quad a = R \alpha$$

$$a = \frac{x m g}{(m + M / 2) l}$$